23MID0233

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In [3]:
```
import pandas as pd
```

A

Create a simple series of Student_Names and Programmes they belong with

In [11]:
```
simple_srs=pd.Series(["santhosh","santhu","santo"],index=["CSE","ECE","MECH_ENG"])
print(simple_srs)
```

```
CSE         santhosh
ECE           santhu
MECH_ENG       santo
dtype: object
```

B

Create a series from a list, tuple, dictionary and a custom index

In [14]:
```
#list
lst=[1,2,3,4,5]
s=pd.Series(lst)
print(s)
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
```

In [15]:
```
#tuple
tple=(2,3,4,5,6)
s1=pd.Series(tple)
print(tple)
```

```
(2, 3, 4, 5, 6)
```

In [16]:
```
#dictionary
dictionary={
    "santo":23,
    "turly":54,
    "tampop":76,
    "meme":10
}
s2=pd.Series(dictionary)
print(s2)
```

```
santo     23
turly     54
tampop    76
meme      10
dtype: int64
```

In [17]:
```
#custom index
custom=pd.Series(["santhosh","santhu","santo"],index=[21,20,19])
print(custom)
```

```
21    santhosh
20      santhu
19       santo
dtype: object
```

C

Create two series with CAT I and CAT II marks of students and explore element-wise

operations

In [27]:
```
cat1=pd.Series([23,45,34,33,38])
cat2=pd.Series([40,42,45,41,47])
print(cat1 + cat2)
```

```
0    63
1    87
2    79
3    74
4    85
dtype: int64
```

D

Write conditions to check if data in a series are integers/chars/Not a Number

In [41]:
```
import numpy as np
data = pd.Series([10, "Santhosh", 25, "CSE", np.nan, 50])

print("\nD) Data:")
print(data)


# Check each value
for value in data:

    if pd.isna(value):
        print(value, "-> Not a Number (NaN)")

    elif isinstance(value, int):
        print(value, "-> Integer")
```

```
    elif isinstance(value, str):
        print(value, "-> Character/String")

    else:
        print(value, "-> Other")
```

```
D) Data:
0          10
1    Santhosh
2          25
3         CSE
4         NaN
5          50
dtype: object
10 -> Integer
Santhosh -> Character/String
25 -> Integer
CSE -> Character/String
nan -> Not a Number (NaN)
50 -> Integer
```

E

Create a series for Product Names, Prices, Quantities in a retail store

In [34]:
```
index=["product_name","price","quantity","exp_date"]
product_data=pd.Series(["Oats",300,"500mg",12-2027],index=index)
print(product_data)
```

```
product_name     Oats
price             300
quantity        500mg
exp_date        -2015
dtype: object
```

F

Create a Multilevel series with arrays, tuple, product and dataframe

In [35]:
```
students = ["Santhosh", "Santhosh", "Soumya", "Soumya"]
subjects = ["Python", "Java", "Python", "Java"]

index1 = pd.MultiIndex.from_arrays(
    [students, subjects]
)

marks1 = pd.Series(
    [90, 85, 95, 88],
    index=index1
)

print("\nF) Multilevel Series using Arrays:")
print(marks1)


# Multilevel Series

index2 = pd.MultiIndex.from_tuples([
    ("Santhosh", "Python"),
    ("Santhosh", "Java"),
    ("Soumya", "Python"),
    ("Soumya", "Java")
])

marks2 = pd.Series(
    [90, 85, 95, 88],
    index=index2
)

print("\nMultilevel Series using Tuples:")
print(marks2)


#Multilevel Series using Product

students2 = ["Santhosh", "Soumya"]
subjects2 = ["Python", "Java"]
```

```
index3 = pd.MultiIndex.from_product(
    [students2, subjects2]
)

marks3 = pd.Series(
    [90, 85, 95, 88],
    index=index3
)

print("\nMultilevel Series using Product:")
print(marks3)


#Multilevel DataFrame

data_frame = pd.DataFrame({
    "Student": ["Santhosh", "Santhosh", "Soumya", "Soumya"],
    "Subject": ["Python", "Java", "Python", "Java"],
    "Marks": [90, 85, 95, 88]
})

print("\nMultilevel DataFrame:")
print(data_frame)


# Convert Student and Subject into index
data_frame = data_frame.set_index(["Student", "Subject"])

print("\nDataFrame with Multilevel Index:")
print(data_frame)
```

```
F) Multilevel Series using Arrays:
Santhosh  Python    90
          Java      85
Soumya    Python    95
          Java      88
dtype: int64

Multilevel Series using Tuples:
Santhosh  Python    90
          Java      85
Soumya    Python    95
          Java      88
dtype: int64

Multilevel Series using Product:
Santhosh  Python    90
          Java      85
Soumya    Python    95
          Java      88
dtype: int64

Multilevel DataFrame:
    Student Subject  Marks
0  Santhosh  Python     90
1  Santhosh    Java     85
2    Soumya  Python     95
3    Soumya    Java     88

DataFrame with Multilevel Index:
                  Marks
Student  Subject
Santhosh Python      90
         Java        85
Soumya   Python      95
         Java        88
```

G

Access the data elements using First Level Index and Second Level Index.

In [37]:
```
print("First Level Index - Santhosh:")
print(marks3["Santhosh"])


print("\nSecond Level Index - Python:")
print(marks3[:, "Python"])


# Access one value using both levels
print("\nSanthosh's Python Mark:")
print(marks3["Santhosh", "Python"])
```

```
First Level Index - Santhosh:
Python    90
Java      85
dtype: int64

Second Level Index - Python:
Santhosh    90
Soumya      95
dtype: int64

Santhosh's Python Mark:
90
```

H

Access a specific data item using .loc

In [39]:
```
print(marks3.loc[("Santhosh", "Python")])


# Access all marks
print("\nAll marks of Santhosh:")
print(marks3.loc["Santhosh"])
```

```
# Access Python marks of all students
print("\nPython marks of all students:")
print(marks3.loc[:, "Python"])
```

```
90

All marks of Santhosh:
Python    90
Java      85
dtype: int64

Python marks of all students:
Santhosh    90
Soumya      95
dtype: int64
```

In [ ]: